Umar Hassan Makki Mohammed

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EDUCATION

Northeastern University, Boston, MA *Sep 2025 – Apr 2027*
Master of Science in Robotics; GPA: 3.98
Relevant Coursework: Robot Mechanics and Control; Mobile Robotics; Robot Sensing and Navigation; Reinforcement Learning; Autonomous Field Robotics; Machine Learning and Small Data

KL University, India *Aug 2021 – May 2025*
Bachelor of Technology in Artificial Intelligence and Data Science

EXPERIENCE

Teaching Assistant, Robotics: Science and Systems (CS 4610) *Sep 2026 – Present*
Northeastern University, Boston, MA (Instructor: Prof. Zhi Tan)
- Lead weekly TA sessions and office hours for a class of 30+ students, guiding them one on one through forward and inverse kinematics, trajectory planning, and control, and debugging their code and simulation setups.
- Support the instructor with semester course planning and mentor teams on the term long robotics project, grading code and deliverables against a uniform rubric with detailed technical feedback.

PROJECTS

GPS Denied Autonomous Drone Navigation in Confined Environments *Jan 2026 – Apr 2026*
- Developed a fully autonomous UAV stack for GPS denied tunnel networks, integrating LIO-SAM LiDAR inertial SLAM (RTAB-Map fallback for featureless corridors), OctoMap 3D occupancy mapping, and Nav2 frontier-based exploration, achieving 100% waypoint coverage (114/114 goals).
- Achieved globally consistent, drift-free localization via Scan Context loop closure (4 to 6 constraint edges per closure), benchmarked against ground truth using ATE/RPE metrics in the evo toolbox.

Multi-View ACT UR5e Servoing: Autonomous Cable Insertion *2026*
Intrinsic AI for Industry Challenge 2026 (Qualification Phase)
- Built a three-tier hybrid manipulation pipeline (ACT imitation policy, multi-view perception, and closed loop visual servoing) for autonomous fiber optic cable insertion (~0.5 to 1.0 mm clearance) on a UR5e arm, trained on 302 teleoperated episodes.
- Engineered a YOLOv8s-OBB + PnP 6-DoF pose estimator with synthetic auto-labeling (5,160 labels in under an hour, mAP50 ~ 0.965), plus multi-view scene registration and a hybrid visual servo with failure recovery.

Curiosity Driven Exploration for Sparse Rewards: Reinforcement Learning *Jan 2026 – Apr 2026*
- Implemented PPO+RND, DQN+ICM, and Rainbow DQN to tackle exploration in Adventure 2600, one of Atari's most notoriously unsolved sparse-reward games, training over 120M steps across three independent implementations.
- Engineered RAM based state extraction, a unified dense reward-shaping framework, and a curriculum learning pipeline; diagnosed and patched recurring failure modes (greedy policy collapse, reward-farming exploits, incorrect ALE RAM addresses), improving gold-key pickup 39x over baseline through action-space pruning.

PGTP-Net (v1): Pose-Guided Pedestrian Trajectory Prediction *Sep 2025 – Dec 2025*
- Built a dual-stream architecture (Social-LSTM with attention pooling plus a Transformer pose encoder) fused into a CVAE decoder that generates K = 20 candidate 4.8-second futures conditioned on 2D/3D pose keypoints.
- Reduced trajectory error (ADE/FDE) by 35 to 46% on JTA with 3D ground-truth poses and 25 to 28% on ETH/UCY with estimated 2D poses, validating that skeletal pose carries anticipatory signal point-mass models discard.

LEADERSHIP & ACTIVITIES

Competitions Coordinator, Business Operations *Jun 2026 – Present*
Students for the Exploration and Development of Space (SEDS), Northeastern University
- Coordinate competitions for every team across the SEDS, owning logistics, scheduling, and communication between competition organizers and members to keep teams aligned and on track.

Member: NU Rover (autonomous rover team), AeroNU (aerospace & rocketry), Astronomy Club, NUHOC (Northeastern University Hiking Outing Club)

TECHNICAL SKILLS

Programming: Python (PyTorch, TensorFlow), C++, C, CUDA, ROS, ROS2, Linux, Git, Docker
Hardware & Simulation: UR5e arm, Intel RealSense D435i, Velodyne VLP-16 LiDAR, Dynamixel servos, IMU, Gazebo, RViz, SolidWorks, Fusion 360, Onshape
SLAM, Planning & Control: LIO-SAM, RTAB Map, OctoMap, Kalman filtering, EKF, Nav2, frontier based exploration, trajectory planning, inverse kinematics (DLS), Model Predictive Control
Perception & Manipulation: 6-DoF pose estimation (PnP, SMPL), YOLO / YOLO-OBB, visual servoing, multi view scene registration, camera calibration, depth perception
AI & Deep Learning: Reinforcement learning (PPO, RND, ICM, Rainbow DQN), Action Chunking Transformers (LeRobot ACT), Transformers, CVAE, LSTM, ResNet